Yuhao Yang

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Age: 23 Nationality: Mainland China Gender: Male First language: Mandarin English Speaking: Advanced

EDUCATION

Beihang University, School of Electronic Information Engineering **Beijing, China**

B.Eng. in **Communication Engineering** (ShanghaiRanking Top 7% in China by subject) *Sept.2018-June.2022*

University of Leeds, School of Electronic & Electrical Engineering **Leeds, United Kingdom**

MSc(Eng). in **Electronic and Electrical Engineering** (QS Top 100 by Subject) - Pass with Distinction *Sept.2022-Aug.2023*

Hard SKILLS

Programming Skills: Python, C/C++, Verilog, MATLAB.

Large Language Model: Continue Pretrain, Supervised Fine-tuning (Full/LoRA), Quantization and Deployment, Prompt Engineering, Embodied AI.

Development Skills: Backend DevOps, Linux shell, git, docker, algorithm testing, project reproduction and optimization.

IELTS: 7.5 **TOEFL (iBT):** 109 **GRE:** V156+Q168+AW3.5

EXPERIENCE

Centre for Artificial Intelligence and Robotics, Chinese Academy of Sciences **Hong Kong**

Algorithm Engineer (Large Language Model), *Sept.2023-Present*

- Conducted the whole process of Continue-Pretraining and Supervised Fine-Tuning (Full SFT, LoRA and QLoRA) on several Large Language Models on different tasks, including LLaMa 2 7/13/70b, Mixtral of experts 8x7b, InternLM2, etc. Able to utilize distributed training frameworks such as Megatron, Accelerate, DeepSpeed, trl, and others like Peft, QLoRA, etc. Own a deep understanding of the hyper parameters during training and fine-tuning through dozens of training processes.
- Responsible for accelerating inference deployment of large models. Through low-precision hybrid quantization, operator optimization, memory access optimization and other technologies, the video memory usage is reduced to 20% with almost no loss of accuracy, and the inference generation speed is increased to 200%.
- Eliminated illusions and optimized the experience of the LLM application. Added knowledge bases with multi-scale and multi-channel recall for retrieval augmented generation (RAG), and support multi-round dialogue based on documents, voice input, image recognition, segmentation and other visual plugin calling functions.

Cambricon Technologies **Beijing**

Algorithm Research Intern, *Dec.2021-April.2022*

- Facilitated a raw video de-noise algorithm development and tried different methods to train the de-noise model.
- Acquired cutting-edge knowledge in designing deep learning acceleration hardware.
- Enhanced my knowledge in CUDA PTX and wrote a tutorial document with over a hundred instructions.

RESEARCH EXPERIENCE

Project: The Development of EMVS Computing Accelerator based on FPGA

Research Assistant, ARCHIP Lab, Supervisor: Prof: Jianlei Yang *Feb.2021-June.2021*

- Implemented a high-parallelism, high-throughput pipeline structure to accelerate key operations in EMVS algorithm.
- Proposed the architecture design and verification, together with the IC design of the hardware part.
- Achieved up to 24× improvement in efficiency compared with Intel i5 CPU platform. Work published in top conference.

  Eventor: An Efficient Event-Based Monocular Multi-View Stereo Accelerator on FPGA Platform, ACM/IEEE DAC 2022.

OTHERS

China International College Students' Innovation and Entrepreneurship Competition - Second Prize *June.2021*

President of Beihang Skiing Society & Captain of the Snowboard team *Sept.2021 – July.2022*

Kaggle - Google-Fast or Slow? Predict AI Model Runtime - Silver (Top 3%) [redacted]

Kaggle - Child Mind Institute-Detect Sleep States - Silver (Top 2%)